

Intuitively, $(\widehat{M}_\lambda^b)_{1 \leq b \leq B+1}$ simulate values of the MMD test statistic under the null hypothesis $\mathcal{H}_0: p = q$, the quantile $\widehat{q}_{1-\alpha}^{\lambda, B}$ is defined such that only an α -proportion of the simulated test statistics are greater than $\widehat{q}_{1-\alpha}^{\lambda, B}$. As such, as shown in Proposition 1, under $\mathcal{H}_0$, the probability that the MMD test statistic is greater than the quantile (*i.e.* rejecting the null) is non-asymptotically at most α .

As shown in Appendix E.1, the p -value of the test can be computed as

$$p_{\text{val}}^\lambda := \frac{1}{B+1} \left(1 + \sum_{b=1}^B \mathbb{1}(\widehat{M}_\lambda^b \geq \widehat{M}_\lambda^{B+1}) \right)$$

and satisfies the property that

$$p_{\text{val}}^\lambda \leq \alpha \iff \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) > \widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n).$$

Note that computing the quantile $\widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n)$ requires sorting the simulated test statistics, while computing the p -value p_{val}^λ does not.

We now prove that this single MMD test has the desired non-asymptotic level α , which we stress differs from the original asymptotic MMD test of Gretton et al. (2012a). We believe that this non-asymptotic property will contribute to the wide use of those MMD-based tests.

Proposition 1 (proof in Appendix E.1) *For fixed bandwidth $\lambda \in (0, \infty)^d$, $\alpha \in (0, 1)$ and $B \in \mathbb{N} \setminus \{0\}$, the test $\Delta_\alpha^{\lambda, B}$ has non-asymptotic level α , that is*

$$\mathbb{P}_{p \times p \times r} \left(\Delta_\alpha^{\lambda, B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 1 \right) \leq \alpha$$

for all probability density functions p on $\mathbb{R}^d$.

This single MMD test $\Delta_\alpha^{\lambda, B}$ depends on the choice of bandwidth λ . In practice, one would like to choose λ such that the test $\Delta_\alpha^{\lambda, B}$ has high power against most alternatives. In general, a smaller bandwidth gives a narrower kernel which is well suited to detect local differences between probability densities such as small perturbations. On the other hand, a larger bandwidth gives a wider kernel which is better at detecting global differences between probability densities. We verify those intuitions in our experiments presented in Section 5. While insightful, those do not tell us exactly how to choose the bandwidth.

As mentioned in the introduction, in practice, there exist two common approaches to choosing the bandwidth of the single MMD test. The first one, proposed by Gretton et al. (2012a), is to set the bandwidth to be equal to the median inter-sample distance. The second approach involves splitting the data into two parts where the first half is used to choose the bandwidth that maximises the asymptotic power, and the second half is used to run the test. This was initially proposed by Gretton et al. (2012b) for the linear-time MMD estimator, and later generalised by Liu et al. (2020) to the case of the quadratic-time MMD estimator. The former approach has no theoretical guarantees, while the latter can suffer from a loss of power caused by the use of less data to run the test. Those two methods are further analysed in our experiments in Section 5.

In Sections 3.3 and 3.4, we obtain theoretical guarantees for the power of the single MMD test $\Delta_\alpha^{\lambda, B}$ and specify the choice of the bandwidth that leads to minimax optimality.

3.3 Controlling the power of the single MMD test

We start by presenting conditions on the discrepancy measures $\text{MMD}_\lambda(p, q)$ and $\|p - q\|_2$ under which the probability of type II error of the single MMD test

$$\mathbb{P}_{p \times q \times r}(\Delta_\alpha^{\lambda, B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 0) = \mathbb{P}_{p \times q \times r}(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \leq \widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n))$$

is controlled by a small positive constant β . We then express these conditions in terms of the bandwidth λ . We find a sufficient condition on the value of $\text{MMD}_\lambda^2(p, q)$ which guarantees that the single MMD test $\Delta_\alpha^{\lambda, B}$ has power at least $1 - \beta$ against the alternative $\mathcal{H}_a: p \neq q$.

Lemma 2 (proof in Appendix E.2) *For $\alpha, \beta \in (0, 1)$, and $B \in \mathbb{N} \setminus \{0\}$, the condition*

$$\mathbb{P}_{p \times q \times r}(\text{MMD}_\lambda^2(p, q) \geq \sqrt{\frac{2}{\beta} \text{var}_{p \times q}(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n))} + \widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n)) \geq 1 - \frac{\beta}{2}$$

is sufficient to control the probability of type II error such that

$$\mathbb{P}_{p \times q \times r}(\Delta_\alpha^{\lambda, B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 0) \leq \beta.$$

If the densities p and q differ significantly in the sense that $\text{MMD}_\lambda^2(p, q)$ satisfies the condition of Lemma 2, then the probability of type II error of the single MMD test $\Delta_\alpha^{\lambda, B}$ against that alternative hypothesis is upper-bounded by β . The condition includes two terms: the first term depends on β as well as on the variance of $\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)$, and the second is the conditional quantile estimated using the Monte Carlo method with either permutations or a wild bootstrap. In the next two propositions, we make this condition more concrete by providing upper bounds for the variance and the estimated conditional quantile. In particular, the upper bounds are expressed in terms of the bandwidth λ and the sample sizes m and n , which guides us towards the choice of the bandwidth with an optimal guarantee. We start with the variance term.

Proposition 3 (proof in Appendix E.3) *Assume that $\max(\|p\|_\infty, \|q\|_\infty) \leq M$ for some $M > 0$. Given φ_λ as defined in Equation (9) and $\psi := p - q$, there exists a positive constant $C_1(M, d)$ such that*

$$\text{var}_{p \times q}(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)) \leq C_1(M, d) \left(\frac{\|\psi * \varphi_\lambda\|_2^2}{m + n} + \frac{1}{(m + n)^2 \lambda_1 \cdots \lambda_d} \right).$$

We now upper bound the estimated conditional quantile $\widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n)$ in terms of λ and $m + n$. Since this is a random variable, we provide a bound which holds with high probability.

Proposition 4 (proof in Appendix E.4) *We assume $\max(\|p\|_\infty, \|q\|_\infty) \leq M$ for some $M > 0$, $\alpha \in (0, 0.5)$ and $\delta \in (0, 1)$. For all $B \in \mathbb{N}$ satisfying $B \geq \frac{3}{\alpha^2}(\ln(\frac{4}{\delta}) + \alpha(1 - \alpha))$, we have*

$$\mathbb{P}_{p \times q \times r}(\widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \leq C_2(M, d) \frac{1}{\sqrt{\delta}(m + n)} \frac{\ln(\frac{1}{\alpha})}{\sqrt{\lambda_1 \cdots \lambda_d}}) \geq 1 - \delta$$

for some positive constant $C_2(M, d)$.

Note that while this bound looks similar to the one proposed by Albert et al. (2022, Proposition 3) for independence testing, it differs in two major aspects. Firstly, while they consider the theoretical (unknown) quantile $q_{1-\alpha}^\lambda$, we stress that our bound holds for the random variable $\widehat{q}_{1-\alpha}^{\lambda,B}(\mathbb{Z}_B|\mathbb{X}_m, \mathbb{Y}_n)$, which is the conditional quantile estimated using the Monte Carlo method with either permutations or a wild bootstrap. Secondly, our bound holds for any bandwidth $\lambda \in (0, \infty)^d$ without any additional assumptions. In particular, we do not require the restrictive condition that $(m+n)\sqrt{\lambda_1 \cdots \lambda_d} > \ln(\frac{1}{\alpha})$ which can in some cases imply that the sample sizes need to be very large.

Having obtained upper bounds for $\text{var}_{p \times q}(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n))$ and $\widehat{q}_{1-\alpha}^{\lambda,B}(\mathbb{Z}_B|\mathbb{X}_m, \mathbb{Y}_n)$, we now combine these with Lemma 2 to obtain a more concrete condition for type II error control. More specifically, the refined condition depends on λ , $m+n$ and β , and guarantees that the probability of type II error of the single MMD test $\Delta_\alpha^{\lambda,B}$, against the alternative (p, q) defined in terms of the L^2 -norm, is at most β .

Theorem 5 (proof in Appendix E.5) *We assume $\max(\|p\|_\infty, \|q\|_\infty) \leq M$ for some $M > 0$, $\alpha \in (0, e^{-1})$, $\beta \in (0, 1)$ and $B \in \mathbb{N}$ which satisfy $B \geq \frac{3}{\alpha^2}(\ln(\frac{8}{\beta}) + \alpha(1-\alpha))$. We consider φ_λ as defined in Equation (9) and let $\psi := p - q$. Assume that $\lambda_1 \cdots \lambda_d \leq 1$. There exists a positive constant $C_3(M, d)$ such that if*

$$\|\psi\|_2^2 \geq \|\psi - \psi * \varphi_\lambda\|_2^2 + C_3(M, d) \frac{\ln(\frac{1}{\alpha})}{\beta(m+n)\sqrt{\lambda_1 \cdots \lambda_d}},$$

then the probability of type II error satisfies

$$\mathbb{P}_{p \times q \times r}(\Delta_\alpha^{\lambda,B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 0) \leq \beta.$$

The main condition of Theorem 5 requires $\|p - q\|_2^2$ to be greater than the sum of two quantities. The first one is the bias term $\|\psi - \psi * \varphi_\lambda\|_2^2$ and the second one comes from the upper bounds in Propositions 3 and 4 on the variance $\text{var}_{p \times q}(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n))$ and on the estimated conditional quantile $\widehat{q}_{1-\alpha}^{\lambda,B}(\mathbb{Z}_B|\mathbb{X}_m, \mathbb{Y}_n)$. Now, we want to express the bias term $\|\psi - \psi * \varphi_\lambda\|_2^2$ explicitly in terms of the bandwidths λ . For this, we need some smoothness assumption on the difference of the probability densities.

3.4 Uniform separation rate of the single MMD test over a Sobolev ball

We now assume that $\psi := p - q$ belongs to the Sobolev ball $\mathcal{S}_d^s(R)$ defined in Equation (1). This assumption allows us to derive an upper bound on the uniform separation rate of the single MMD test in terms of the bandwidth λ and of the sum of sample sizes $m+n$.

Theorem 6 (proof in Appendix E.6) *We assume that $\alpha \in (0, e^{-1})$, $\beta \in (0, 1)$, $s > 0$, $R > 0$, $M > 0$ and $B \in \mathbb{N}$ satisfying $B \geq \frac{3}{\alpha^2}(\ln(\frac{8}{\beta}) + \alpha(1-\alpha))$. Given that $\lambda_1 \cdots \lambda_d \leq 1$, the uniform separation rate of the test $\Delta_\alpha^{\lambda,B}$ over the Sobolev ball $\mathcal{S}_d^s(R)$ can be upper bounded as follows*

$$\rho(\Delta_\alpha^{\lambda,B}, \mathcal{S}_d^s(R), \beta, M)^2 \leq C_4(M, d, s, R, \beta) \left(\sum_{i=1}^d \lambda_i^{2s} + \frac{\ln(\frac{1}{\alpha})}{(m+n)\sqrt{\lambda_1 \cdots \lambda_d}} \right)$$

for some positive constant $C_4(M, d, s, R, \beta)$.